Python Engineering - Reference Guide

Class 07 - NumPy Arrays: quick guide

Initial reference for the functions and concepts used in EX15. It is intentionally limited to the NumPy foundation we need now.

1. What is an ndarray?

An **ndarray** is NumPy's main numerical data structure. It resembles a list but is optimized for numerical data and vectorized operations.

```
import numpy as np

a = np.array([1, 2, 3])
```

2. Essential properties

Property	Meaning	Example
.shape	size along each dimension	(103, 3)
.ndim	number of dimensions	2
.size	total number of elements	309
.dtype	stored numeric type	float64

3. 2D indexing and slicing

```
A[row, column]

A[0, :]     # first row
A[:, 0]     # first column
A[-1, :]    # last row
A[2:5, 1]   # rows 2..4, column 1
```

Mental rule: for a NumPy matrix we normally write row first, then column.

4. Element-wise arithmetic

```
a = np.array([1.0, 2.0, 3.0])

a + 2       # [3,4,5]
a * 2       # [2,4,6]
a**2        # [1,4,9]
np.sqrt(a)  # sqrt of every element
```

5. Vectorization

Vectorization means expressing an operation over complete arrays instead of writing a Python loop for every element.

```
energy = g * heights + 0.5 * velocities**2
```

6. Functions used in EX15

Function	What it does
np.array(data)	convert compatible data into ndarray
np.max(a)	maximum value
np.min(a)	minimum value
np.argmax(a)	index of maximum

Function	What it does
np.argmin(a)	index of minimum
np.sum(a)	sum of elements
np.mean(a)	mean

7. List vs NumPy array

```
[1, 2, 3] * 2
# [1,2,3,1,2,3]

np.array([1,2,3]) * 2
# [2,4,6]
```

8. Extracting columns

```
trajectory_array = np.array(trajectory)
heights = trajectory_array[:, 0]
velocities = trajectory_array[:, 1]
times = trajectory_array[:, 2]
```

9. Common mistakes

Mistake	Idea
mixing row and column	A[:,0] is column 0; A[0,:] is row 0
expecting list behavior	ndarray arithmetic is numerical and element-wise
mixing incompatible types	NumPy tries to find a common dtype
using loops by habit	for numerical work, look for a vectorized expression first

Mini cheat sheet

```
np.array(data)        create array
A.shape               dimensions
A.ndim                number of axes
A[:, 0]               column 0
A[0, :]               row 0
np.max(A)             maximum
np.argmax(A)          max index
A * scalar            element-wise multiply
```